

Question	Answer	Marks
8(a)	force per unit current	B1
	force per unit length (of wire)	B1
	current normal to (magnetic) field	B1
8(b)(i)	forces (on PQ and RS) are horizontal	B1
	(hence they create) no moment about the pivot	B1
	or	
	forces (on PQ and RS) are equal and opposite	(B1)
	(hence there is) no <u>net</u> force (on the two sections)	(B1)
8(b)(ii)	realisation of the need to apply moments	C1
	$BILx = mgy$	C1
	$B \times 2.7 \times 1.2 \times 10^{-2} \times 7.5 = 45 \times 10^{-6} \times 9.81 \times 8.8$	
	$B = 1.6 \times 10^{-2} \text{ T}$	A1

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Question	Answer	Marks
9(c)	One of the following is correct: 5.16	D1